UNITED INTERNATIONAL UNIVERSITY

Department of Computer Science and Engineering (CSE)

40 min

Course : DSA II Lab | **Course Code: CSE 2218**

Name : | **ID:**

Evaluation 4

SN	Questions	Marks
1	Prim's algorithm is a very popular algorithm for determining the Minimum Spanning Tree from a graph. There is also another type of spanning tree called the Maximum Spanning Tree which can also be determined by making slight modifications in the Prim's Algorithm.	10

Now your task is to use Prim's Algorithm to find the **Maximum Spanning Tree** from the given graph. You must take the graph **inputs from the user**. Afterwards print the **key values** and **parent values** for the spanning tree and determine the Maximum Spanning Tree Weight Sum.

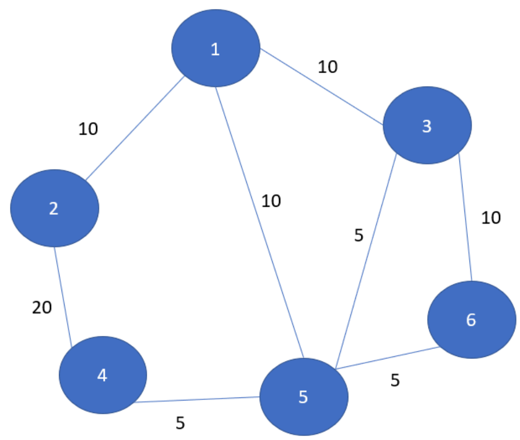

Figure 1 : Sample Input Graph

The Sample Input consists of the number of vertices and the number of edges, followed by the entry of the edges in the format (start vertex, end vertex, weight).
The sample output consists of the key values and the parent values along with the total weight of the maximum spanning tree.

Sample Input	Sample Output
6 8 1 2 10 2 4 20 4 5 5 5 6 5 6 3 10 1 3 10 1 5 10 3 5 5	parent values: -1 1 1 2 1 3 key values: 0 10 10 20 10 10 The weight of the maximum spanning tree is 60

Evaluation Rubrics

Marking Criteria	Marks
Input from the user	3
Prim's Algorithm Modification	5
Overall Correctness	2

SN	Questions	Marks
2	You have been recently elected as the mayor of the city “Aurelia Metropolis”. The city consists of 6 major locations namely Voxis, Strata, Kryton, Eclipta, Nexara and Zenith. Now observe that there are different routes between these city locations as shown in the figure. Each of the routes have a distance associated with them. However, as shown in the graph, not all of those routes are safe. You want to take up a metro rail project connecting all the city locations, however, you want the paths to be safe.	

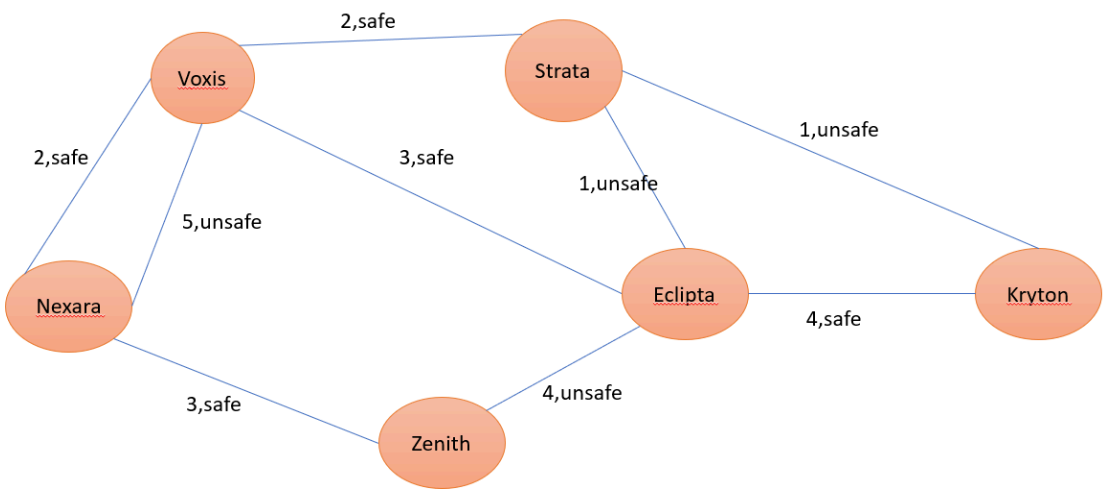

Figure 2 : Aurelia Metropolis – City Routes

Now, your task is to apply Kruskal's algorithm on the above graph (or some other similar graph which will be provided as input by the user). You are required to determine a **safe** minimum spanning tree which will consist of only the safe routes. Print the edges and the weight of the tree as shown in the sample input and output.

Sample Input	Sample Output
Number of cities : 6 Number of routes: 9 Voxis Nexara 2 safe Voxis Nexara 5 unsafe Nexara Zenith 3 safe Voxis Eclipta 3 safe Voxis Strata 2 safe Zenith Eclipta 4 unsafe Strata Eclipta 1 unsafe Eclipta Kryton 4 safe Strata Kryton 1 unsafe	The selected edges: (Voxis,Nexara,2,safe) (Voxis,Strata,2,safe) (Nexara,Zenith,3,safe) (Voxis,Eclipta,3,safe) (Eclipta,Kryton,4,safe) Total weight of the MST : 14

Evaluation Rubrics

Marking Criteria	Marks
Input from the user with town names	3
Kruskal’s Algorithm Adaptation and able to detect safe paths	5
Overall Correctness	2